$$P = I^2 R = [I_0 + i\cos(\omega t - \varphi_0)]^2 R_0(1+\lambda\theta)$$
$$= (I_0^2 + \frac{1}{2}i^2)R_0(1+\lambda\theta) + 2I_0 i R_0(1+\lambda\theta)\cos(\omega t - \varphi_0) + \frac{1}{2}i^2 R_0(1+\lambda\theta)\cos 2(\omega t - \varphi_0)$$